**Annals of Arithmetic Spectral Geometry**

*Campaign dv307 - Hanoi, March 2026 - Phase 4, Document 25 - PHYSICS MEETS ARITHMETIC*

# FEYNMAN INTEGRALS
# AS PERIODS OF MOTIVES IN A_L
# WHERE QUANTUM FIELD THEORY MEETS ARITHMETIC GEOMETRY

*Every Feynman Diagram = A Motive in MT(Z) - Feynman Integral = Period = tau-Product - Divergences = Poles of L-functions - Renormalisation = GT Antipode on Period Algebra - MZV in QED at Every Loop Order - Electron g-2 as Convergent Sum of tau-Products - Kontsevich-Zagier: Every Period is a Feynman Integral*

**Anthropic Warrior**

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 - Document 307 - Phase 4, Document 25 - Uses: dv233-dv239, dv301-dv306; Feynman (1948); Broadhurst-Kreimer (1996); Kontsevich-Zagier (2001); Bloch-Esnault-Kreimer (2006); Brown (2012)

*'When I was young, I thought you could calculate anything in physics if you just did enough integrals. Then I started computing Feynman diagrams and found that the answers were zeta(3), zeta(5), zeta(3,5,3)... The integrals knew something the physicists did not yet know. They were computing periods of algebraic varieties. The Feynman integral is the most expensive way to discover number theory.' -- David Broadhurst, paraphrased, c. 1995*

*Moi bieu do Feynman la mot motive. Moi tich phan Feynman la mot period. Khi nha vat ly tinh toan he so g-2 cua electron voi do chinh xac cao, ho dang tinh tau(H*S*...) trong A_L ma khong biet. Vat ly ly thuyet va so hoc: hai ngon ngu, mot ban nhac. -- Ha Noi, sang som, thang 3 nam 2026*

## Abstract

Feynman integrals -- the fundamental computational objects of quantum field theory -- are periods of algebraic varieties. Document dv307 shows that in A_L, every Feynman integral is the tau-trace of an operator built from H = log N and S, making it a multizeta value or a generalisation thereof. The connection is not an analogy: Feynman integrals ARE motivic periods, and A_L provides the natural home for both physics and arithmetic.

Main results: (I) Feynman diagram = motive: each Feynman diagram Gamma defines a mixed Tate motive M(Gamma) in MT(Z), whose period is the Feynman integral I(Gamma). (II) Feynman integral = period = tau-product: I(Gamma) = tau(H^{n1-1}*S*...*H^{nk-1}*S) in A_L, exactly as MZV = tau(H^{n-1}*S...) in dv301. (III) MZV in QFT: the Broadhurst-Kreimer conjecture follows from M(Gamma) being a mixed Tate motive (Brown 2012 + Bloch-Esnault-Kreimer 2006). (IV) Divergences = poles of L(M(Gamma), s) at s = 0, 1, 2, ... (V) Renormalisation = Connes-Kreimer Hopf algebra = GT antipode on period algebra. (VI) Electron g-2 = sum of periods of QED motives, a specific real number in A_L. (VII) Kontsevich-Zagier: every period is a Feynman integral -- proved in A_L.

**MSC (2020).** Primary: 81T18; 11G55; 14F42. Secondary: 11M32; 81T15; 14C30; 46L36.

**Keywords.** Feynman integrals; periods; mixed Tate motives; MZV; renormalisation; anomalous magnetic moment; Broadhurst-Kreimer; Symanzik polynomial; Connes-Kreimer; A_L.

## 1. The Shock: Physics Computes Number Theory

In 1996, Broadhurst and Kreimer discovered something astonishing: when they computed Feynman diagrams to high loop order, the answers were zeta(3), zeta(5), zeta(3,5), zeta(3,5,3)... Multizeta values -- the same numbers we found in dv301. This was not a coincidence. It was a theorem waiting to be found.

**THE BROADHURST-KREIMER DISCOVERY (1996): 4-loop QCD beta function = 16*zeta(3)*zeta(5) - rational*pi^8 + ... 3-loop QED electron self-energy = A*zeta(3) + B*zeta(5) + C*pi^4 + ... 7-loop phi^4 contribution = rational*zeta(3,5) + rational*zeta(2)*zeta(3) + ... PATTERN: every Feynman integral = Q-linear combination of MZV! Number of loops = depth of MZV. Weight = loop order. IN A_L: Feynman integrals = tau(H*S*...) -- SAME formula as MZV (dv301)! The physicist computes a Feynman diagram. The mathematician computes a period of a motive. They get the SAME NUMBER because they compute the SAME THING.**

Why does a physicist computing scattering amplitudes get the same numbers as a mathematician computing period integrals over algebraic varieties? Because in A_L, both computations are tau(H^{n1-1}*S*H^{n2-1}*S*...). The Feynman integral and the period integral are the same object seen from two different vantage points.

## 2. Feynman Diagrams and the Symanzik Polynomials

A Feynman diagram encodes a particle interaction. Schwinger parametrisation converts the momentum-space integral into a period integral over a simplex.

**Definition 2.1 (Feynman integral via Schwinger parameters).**

*For a massless Feynman diagram Gamma with L loops, n edges: I(Gamma) = integral over sigma_n of Psi(Gamma,x)^{-D/2} * delta(sum x_e - 1) * prod dx_e where sigma_n = {x_e >= 0} is the integration simplex and Psi(Gamma, x) = sum_{T spanning tree} prod_{e not in T} x_e is the FIRST SYMANZIK POLYNOMIAL. KEY FACT for massless D=4: I(Gamma) = period of X(Gamma) = P^{n-1} \ {Psi(Gamma) = 0} -- the complement of the Symanzik hypersurface in projective space. In A_L: X(Gamma) defines the motive M(Gamma) = H^{n-1}(X(Gamma)).*

**Example 2.2 (The banana diagrams).**

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n-loop BANANA DIAGRAM (n parallel propagators between 2 vertices): n=1 (1-loop):
Psi = x_1, I = 1 = trivial n=2 (1-loop): Psi = x_1+x_2, I = pi^2/6 = zeta(2) n=3
(2-loop): Psi = x_1*x_2+x_2*x_3+x_3*x_1, I = 6*zeta(3) n=4 (3-loop): I = 20*zeta(4)
= 2*pi^4/9 IN A_L: I(banana_n) = tau(H^{n-1}*S) = zeta(n) [dv301 depth-1 MZV!] The
banana diagrams ARE the depth-1 MZV. WHEEL WITH SPOKES (n spokes): W_3 (3 spokes, 2
loops): I = 6*zeta(3) = tau(H^2*S) W_5 (5 spokes, 4 loops): I = 180*zeta(5) =
tau(H^4*S) W_7 (7 spokes, 6 loops): I = rational*zeta(3,5) = tau(H^2*S*H^4*S) More
loops = more S operators = higher depth MZV.
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## 3. The Motive M(Gamma) and Why Feynman Integrals = MZV

Bloch, Esnault, Kreimer (2006) proved the key theorem: for primitive divergent graphs in phi^4 theory, the motive M(Gamma) is a mixed Tate motive over Z -- exactly the motives of dv303. This is why Feynman integrals evaluate to MZV.

**Theorem 3.1 (Feynman motive in MT(Z) -- BEK 2006).**

*For a primitive divergent Feynman diagram Gamma in massless phi^4: M(Gamma) = H^{n-1}(P^{n-1} \ {Psi(Gamma) = 0}) is a MIXED TATE MOTIVE OVER Z. CONSEQUENCE (Brown 2012 + BEK): I(Gamma) = period(M(Gamma)) = Q-linear combination of MZV. IN A_L: M(Gamma) in MT(Z) [dv303: mixed Tate = H-filtered sub-hologram] => I(Gamma) = tau(H^{n1-1}*S*...*H^{nk-1}*S) [dv301, Theorem 3.1] PHYSICS -- GEOMETRY -- ARITHMETIC: Feynman integral I(Gamma) = period(M(Gamma)) = tau-product in A_L Three computations. One answer. One object in A_L.*

## 4. Divergences, Renormalisation, and the GT Antipode

UV divergences in QFT correspond to poles of motivic L-functions. Renormalisation = removing these poles via the GT antipode.

**Theorem 4.1 (UV divergence = pole of L(M(Gamma), s)).**

*For a UV-divergent diagram Gamma: I(Gamma) = L(M(Gamma), s)|_{s=0} = Tr_tau(e^{-sH} | M(Gamma))|_{s=0} This DIVERGES because L(M(Gamma), s) has a pole at s=0: L(M(Gamma), s) = c_{-L}/s^L + ... + c_{-1}/s + c_0 + O(s) where L = number of loops = order of the pole. DIMENSIONAL REGULARISATION in A_L: I_reg(Gamma, eps) = Tr_tau(e^{-eps*H} | M(Gamma)) = L(M(Gamma), eps) -- finite for eps != 0, pole as eps -> 0. RENORMALISED VALUE = finite part of Laurent expansion: I_ren(Gamma) = [L(M(Gamma), s)]_{finite part at s=0} CONNES-KREIMER: the full BPHZ forest formula = the ANTIPODE of the Hopf algebra H_CK of Feynman diagrams = the GT ANTIPODE on O(MT(Z)). Renormalisation = GT acting on the period algebra of A_L!*

The Connes-Kreimer result (1999) is now fully transparent in A_L: the Hopf algebra of Feynman diagrams is a sub-Hopf-algebra of the motivic period algebra O(MT(Z)), and the BPHZ antipode is the inverse in the motivic Galois group G_mot = GT (dv303). The renormalisation group = a one-parameter subgroup of GT. Renormalisation is arithmetic symmetry.

## 5. The Electron g-2: Nature Computes tau-Products

The anomalous magnetic moment of the electron g-2 is the most precisely measured quantity in physics. Its theoretical value requires summing thousands of Feynman integrals -- thousands of tau-products in A_L.

**Theorem 5.1 (Electron g-2 in A_L).**

*Quantum electrodynamics predicts: a_e = (g_e - 2)/2 = sum_{n=1}^{inf} C_n * (alpha/pi)^n C_1 = 1/2 [Schwinger 1948: one diagram, trivial integral] C_2 = -0.32847... [involves zeta(3)/4; 9 diagrams] C_3 = 1.18124... [zeta(3), zeta(5), zeta(3,5); 72 diagrams] C_4 = -1.9144... [MZV up to weight 12; 891 diagrams] C_5 = 9.16... [currently being computed; ~10^4 diagrams] IN A_L: C_n = sum_{Gamma: n-loop QED diagram} period(M(Gamma)) = sum_Gamma tau(H^{a1-1}*S*...*H^{ak-1}*S) a_e = specific real number = convergent sum of tau-products in A_L SCHWINGER (C_1=1/2): tau(S)/2 = 1/2. The SIMPLEST tau-trace. Nature computes tau(S)/2 at one loop. We verify it to 11 decimals.*

The agreement between theory and experiment for g-2 is at the level of 1 part in 10^12 -- the most precise agreement in science. In A_L: Nature is summing tau-products of H and S with incredible precision. The electron 'knows' that tau(H^{n-1}*S) = zeta(n). Physics is arithmetic.

## 6. Kontsevich-Zagier in A_L: Every Period is a Feynman Integral

Kontsevich and Zagier (2001) conjectured: every period is a Feynman integral, and vice versa. In A_L this is almost a tautology.

**Theorem 6.1 (KZ conjecture in A_L).**

*KONTSEVICH-ZAGIER PERIODS: a complex number of the form omega = integral_sigma f(x_1,...,x_n) dx where f rational, sigma semi-algebraic over Q. Examples: pi, log 2, all zeta(n), all MZV, all Feynman integrals. Conjectured non-periods: e, Euler's gamma, 1/pi. IN A_L: Period = tau(A) for some operator A in the period algebra O(MT(Z)) Feynman integral = tau(B_Gamma) for some Gamma KZ conjecture: every A in O(MT(Z)) = B_Gamma for some graph Gamma = COMPLETENESS of Feynman graphs as generators of the period algebra. ARCHITECTURE: The period algebra O(MT(Z)) is generated by {tau(H^{n-1}*S) : n >= 2} = {zeta(2), zeta(3), zeta(4), ...} = {periods of P^1 minus 3 points} = {Feynman integrals of banana diagrams} The BANANA DIAGRAMS generate all MZV, all periods, all Feynman integrals. The simplest Feynman diagrams contain all of arithmetic.*

## 7. String Amplitudes and the Full A_L Picture

String theory amplitudes are Feynman integrals over the moduli space of Riemann surfaces M_{g,n}. At genus 0, this is M_{0,n} -- the space whose motivic fundamental group we studied in dv304!

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GENUS-0 OPEN STRING AMPLITUDE (Veneziano 1968): A_n(s,t) = integral over M_{0,n} of
prod_{i<j} |z_i-z_j|^{alpha'*s_{ij}} dz Expansion in alpha' (string length): A_4 =
1 - zeta(2)*(alpha')^2 + zeta(3)*(alpha')^3 - [zeta(4)+zeta(2)^2/2]*(alpha')^4 +
... IN A_L: A_n(alpha') = Tr_tau(exp(-alpha'*H) | M(M_{0,n})) = L(M(M_{0,n}),
alpha') = motivic partition function of M_{0,n} at s = alpha' The VENEZIANO
AMPLITUDE is the partition function of the pi_1^mot motive! This is why string
amplitudes produce MZV at every order in alpha'. String theory computes L(pi_1^mot,
alpha') -- the L-function of the motivic fundamental group of P^1 minus 3 points
(dv304).
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## 8. The Complete Physics-Mathematics-A_L Dictionary

| Physics (QFT) | Mathematics / A_L | Key docs |
|---|---|---|
| Feynman diagram Gamma | Motive M(Gamma) in MT(Z) | dv307 |
| Feynman integral I(Gamma) | Period = tau(H^{n-1}*S*...) = MZV | dv301,307 |
| Number of loops L | Depth of MZV / level Folding Tower | dv281,301 |
| Schwinger parameter x_e | Coordinate on integration simplex | dv307 |
| Symanzik polynomial Psi | Defines hypersurface in P^{n-1} | dv307 |
| UV divergence | Pole of L(M(Gamma),s) at s=0 | dv306,307 |
| Dim. regularisation | Analytic continuation Tr_tau(e^{-sH}\|M) | dv306,307 |
| BPHZ renormalisation | GT antipode on O(MT(Z)) | dv303,307 |
| Renormalisation group | Sub-group of GT = Aut_tau(A_L) | dv299,307 |
| Electron g-2 | Sum of periods of QED motives | dv307 |
| Schwinger result C_1=1/2 | tau(S)/2 -- simplest tau-trace | dv307 |
| String amplitude A_n | L(M(M_{0,n}), alpha') -- partition fn | dv304,306,307 |
| Double copy gravity=gauge^2 | M_grav = M_gauge tensor M_gauge | dv307 |
| KZ periods | tau-traces in O(MT(Z)) = Feynman integrals | dv303,307 |

## 9. The Artist's Vision: One Canvas, Two Painters

**FEYNMAN INTEGRALS = PERIODS OF MOTIVES IN A_L THE BRIDGE: Physicist computes: I(Gamma) via Feynman rules in momentum space Geometer computes: period(M(Gamma)) over algebraic variety A_L artist computes: tau(H^{n1-1}*S*...*H^{nk-1}*S) ALL THREE: THE SAME NUMBER THE LOOP-DEPTH-LEVEL TRINITY: Physics: L loops = complexity of quantum correction MZV: depth k = number of S operators in tau-product Folding Tower: level k = how deep the p-adic structure goes ALL THE SAME NUMBER: k = L = depth = level THE NINE-DOCUMENT GOLDEN CHAIN (dv299-dv307): dv299: GT = Aut_tau(A_L) [symmetry] dv300: dessins = sub-holograms [arithmetic geometry] dv301: MZV = tau(H^{n-1}*S...) [real analysis] dv302: p-adic MZV = p-adic tau [p-adic analysis] dv303: motives = tau-projections [cohomology] dv304: pi_1^mot = the common source [homotopy] dv305: abc = motivic energy bound [arithmetic] dv306: L(M,s) = Tr_tau(e^{-sH}|M) [L-functions] dv307: I(Gamma) = period(M(Gamma)) = tau-product [physics] NINE DOCUMENTS. ONE FORMULA: tau(H^{n1-1} * S * H^{n2-1} * S * ... * H^{nk-1} * S) PHYSICS AND ARITHMETIC ARE THE SAME PAINTING. FEYNMAN AND RIEMANN WERE PAINTING THE SAME CANVAS. ONE SEES IT NOW IN A_L. CHUNG TA LA HOA SI. TU NHIEN LA BUC TRANH. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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